

The Data and Database Setup



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NoSQL Databases



Couchbase

MongoDB

Redis

RavenDB

Cassandra

Hbase

DataStax



NoSQL Databases



Key-Value Stores

Document Stores

Column Oriented

Graph Network

Hierarchical

Object-Oriented



MongoDB

Compared to other options MongoDB is relatively simple and is also a popular NoSQL database



Overview



Setup the database

Implement the Flight Plan model

Define the database interface

Implement the database code
- (Parts 1 and 2)



Up Next: The MongoDB Schema



The MongoDB Schema



Flight Plan Data Model

string	FlightPlanId	// Unique Flight Plan Identifier
string	AircraftIdentification	// The tail number of the Aircraft
string	AircraftType	// Make and Model of the Aircraft
int	Airspeed	// The expected airspeed of the flight
int	Altitude	// The altitude the aircraft will maintain
string	FlightType	// Will this be an IFR or a VFR flight
int	FuelHours	// The amount of fuel onboard in hours
int	FuelMinutes	// The minutes of fuel beyond the hour mark
DateTime	DepartureTime	// The time the flight will depart
DateTime	ArrivalTime	// The expected time the flight will arrive
string	DepartureAirport	// The airport from which the flight is departing
string	ArrivalAirport	// The airport at which the flight is arriving
string	Route	// The route the flight will travel
string	Remarks	// Any extra information relevant to the flight
int	NumberOnBoard	// The number of people onboard the flight



The interface must provide a return value that is guaranteed to be available from any database.



Up Next: The Data Model



The Data Model



Up Next: The Database Interface



The Database Interface



Writing an interface to
separate API from database
implementation



Flight Plan API Endpoints

	HTTP VERB	URI Path	Returns Flight Plan	Requires Flight Plan	Requires Flight Plan Id
	GET	http://...			
	GET	http://.../{flight plan id}			
	POST	http://.../file			
	PUT	http://.../{flight plan id}			
	DELETE	http://.../{flight plan id}			



Up Next: MongoDB Implementation



MongoDB Implementation



Coding for clarity



BSON

Binary JSON

Allows more data types that JSON does

Faster to traverse, encode and decode



Where to go
from here



Play with the sample project

- Replace the database
- Add methods to the interface
- Play with the data type



Up Next: Writing the API

